

Grafiche Lama industrial printing	Specifiche cliente: BIOINDUSTRIA L.I.M. Oggetto: foglio illustrativo - MODIFICA nome: AMIODARONE HCL 150mg/3mL (Malaysia) Codice: M2K187/3 Dimensioni: 15 x 39,2 cm. Stampa: F/R - piegato 15x3,3 cm.				Visto si stampi	
	Carattere usato: NOME PRODOTTO: Helv. neue bold corpo 12 Ridim. 95%					
TESTO: Helv neue 7,4 rid. 85% interlinea 7	2a				Data: 25/11/2025	
	Nero			TIPO	Carta	GR/MG
				uso mano		40

AMIODARONE HCL 150mg/3mL INJECTION

BIOINDUSTRIA L.I.M.

Amiodarone Hydrochloride 150mg/3 mL

Presentation:

Amiodarone HCl 150mg/3 mL Injection Bioindustria Laboratorio Italiano Medicinali (L.I.M.):

Each amber neutral glass ampoule contains sterile, clear, colourless or slightly yellowish solution visually free from particulates solution for infusion/injection use. Each carton box contains five 3mL ampoules and one leaflet.

When the product is mixed with diluent solution (1 ampoule in 250ml of 5% dextrose), the mixture is clear and free of particles.

Composition:

Each ampoule contains active ingredient of Amiodarone Hydrochloride 150mg/3mL. Excipients include Benzyl Alcohol 2.02% w/v as preservative, Polysorbate 80, and Water for Injection.

Pharmacodynamic:

Anti-Arrhythmic Properties:

- Prolongation of phase 3 of the cardiac fibre action potential resulting mainly in a decrease in potassium current (class III from Vaughan Williams' classification). This prolongation is not related to heart rate;
- Reduced sinus automaticity leading to bradycardia unresponsive to atropine administration;
- Noncompetitive alpha- and beta-adrenergic inhibition;
- Slowing in sinoatrial, atrial and nodal conduction, which is more marked when the heart rate is high;
- No change in intraventricular conduction;
- Increase in refractory period and decrease of myocardial excitability at the atrial, nodal and ventricular levels;
- Slowing of conduction and prolongation of refractory periods in accessory atrioventricular pathways;
- Reduced cardiac contractility mainly after IV injection.

Pharmacokinetics:

Following IV injection, blood concentration of amiodarone rapidly decreases as impregnation of tissues occurs. The maximum effect is obtained 15 minutes following injection then falls over the next 4 hours. In the absence of repeated injection, the drug is progressively eliminated. Tissue stores are constituted when injections are repeated or therapy is continued orally.

Indications:

Serious rhythm disturbances when the oral route is not appropriate, namely:

- atrial arrhythmia, with rapid ventricular rhythm;
- Wolf-Parkinson-White syndrome tachycardia;
- documented symptomatic and incapacitating ventricular arrhythmia.

Amiodarone Injection is to be used where a rapid response required or where oral administration is not possible.

Cardiopulmonary resuscitation in the event of cardiac arrest related to ventricular fibrillation resistant to external electric shock.

Dosage and Administration:

Amiodarone Injection should only be used when facilities exist for cardiac monitoring, defibrillation, and cardiac pacing.

Amiodarone Injection may be used prior to DC cardioversion.

IV Infusion:

Loading Dose:

The standard recommended dose is 5mg/kg body weight administered in 250mL of 5% dextrose exclusively over a period of 20 minutes to 2 hours. The dose may be repeated 2 – 3 times in 24 hours. The rate of infusion should be adjusted on the basis of clinical response.

Therapeutic effect appears within the first minutes and then progressively decreases. Therefore, an infusion should be set up to take over.

Maintenance Dosage:

10 – 20 mg/kg body weight/ 24 hours (usually 600 – 800mg/ 24 hours and up to 1200mg/ 24 hours) in 250mL of 5% dextrose over a few days.

As soon as an adequate response has been obtained, oral therapy should be initiated concomitantly at the usual loading dose (i.e. 200mg three times a day). Amiodarone Injection should then be phase out gradually.

IV Injection (see Warnings and Precautions):

In extreme clinical emergency the drug may, at the discretion of the clinician, be given as a slow injection of 150-300mg (5mg/kg body weight) in 10-20ml 5% dextrose over a minimum of 3 minutes. This should not be repeated for at least 15 minutes. Patients treated in this way must be closely monitored, e.g. in an intensive care unit.

Cardiopulmonary Resuscitation:

As regards the administration route and in view of the situation in which this indication applies, the use of a central venous catheter is recommended if immediately available; otherwise, the medicinal product may be administered via the peripheral venous route using the largest peripheral vein with the highest flow possible.

The initial intravenous dose is 300mg (or 5mg/kg body weight) diluted in 20mL of 5% dextrose solution and rapidly injected. Additional intravenous administration of 150mg (or 2.5mg/kg body weight) may be envisaged if ventricular fibrillation persists. Do not add any other products to the syringe.

Contraindications:

- Sinus bradycardia and sino-atrial heart block. In patients with severe conduction disturbances (high grade AV block, bifascicular or trifascicular block) or sinus node disease, Amiodarone Injection should be used only in conjunction with a pacemaker.
- Evidence or history of thyroid dysfunction. Thyroid function tests should be performed where appropriate prior to therapy in all patients.
- Severe respiratory failure, circulatory collapse, or severe arterial hypotension; hypotension, heart failure and cardiomyopathy are also contra-indications when using Amiodarone Injection as a bolus injection.
- Known hypersensitivity to iodine or to amiodarone, or to any of the excipients.
- The combination of Amiodarone Injection with drugs which may induce torsades de pointes is contra-indicated (see section Drug Interactions).
- Amiodarone Injection ampoules contain benzyl alcohol. There have been reports of fatal 'gasping syndrome' in neonates (hypotension, bradycardia and cardiovascular collapse) following the administration of intravenous solution containing this preservative. Amiodarone Injection should not be given to neonates or premature babies unless the rhythm disturbance is life threatening and either resistant to other medication or alternative therapy is deemed inappropriate.
- Pregnancy - except in exceptional circumstances (see section Pregnancy and Lactation).
- Lactation (see section Pregnancy and Lactation).

All these above contra-indications do not apply to the use of Amiodarone Injection for cardiopulmonary resuscitation of shock resistant ventricular fibrillation.

Special Warnings and Precautions:

This product is to be used only by a registered medical practitioner with experience in cardiology.

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age.
Not to be used in neonates.

Amiodarone Injection should only be used in a special care unit under continuous monitoring (ECG and blood pressure).

IV infusion is preferred to bolus due to the haemodynamic effects sometimes associated with rapid injection (see section Adverse Reactions). Circulatory collapse may be precipitated by too rapid administration or overdosage (atropine has been used successfully in such patients presenting with bradycardia).

Repeated or continuous infusion via peripheral veins may lead to injection site reactions (see section Adverse Reactions). When repeated or continuous infusion is anticipated, administration by a central venous catheter is recommended.

When given by infusion Amiodarone Injection may reduce drop size and, if appropriate, adjustments should be made to the rate of infusion.

Anaesthesia (see section Drug Interactions): Before surgery, the anaesthetist should be informed that the patient is taking amiodarone.

Cardiac disorders:

Caution should be exercised in patients with hypotension and decompensated cardiomyopathy and severe heart failure (also see section Contraindications).

Amiodarone has a low pro-arrhythmic effect. Onsets of new arrhythmias or worsening of treated arrhythmias, sometimes fatal, have been reported. It is important, but difficult to differentiate a lack of efficacy of the drug from a proarrhythmic effect, whether or not this is associated with a worsening of the cardiac condition. Proarrhythmic effects generally occur in the context of drug interactions and/or electrolytic disorders (see sections Drug Interactions and Adverse Reactions).

Too high a dosage may lead to severe bradycardia and to conduction disturbances with the appearance of an idioventricular rhythm, particularly in elderly patients or during digitalis therapy. In these circumstances, Amiodarone Injection treatment should be withdrawn. If necessary beta-adrenostimulants or glucagon may be given. Because of the long half-life of amiodarone, if bradycardia is severe and symptomatic the insertion of a pacemaker should be considered.

The pharmacological action of amiodarone induces ECG changes: QT prolongation (related to prolonged repolarisation) with the possible development of U-waves and deformed T-waves; these changes do not reflect toxicity.

Respiratory, thoracic and mediastinal disorders (see section Adverse Reactions):

Very rare cases of interstitial pneumonitis have been reported with intravenous amiodarone. When the diagnosis is suspected, a chest X-ray should be performed. Amiodarone therapy should be re-evaluated since interstitial pneumonitis is generally reversible following early withdrawal of amiodarone, and corticosteroid therapy should be considered (see section Adverse Reactions). Clinical symptoms often resolve within a few weeks followed by slower radiological and lung function improvement. Some patients can deteriorate despite discontinuing Amiodarone Injection. Fatal cases of pulmonary toxicity have been reported.

Very rare cases of severe respiratory complications, sometimes fatal, have been observed usually in the period immediately following surgery (adult acute respiratory distress syndrome); a possible interaction with a high oxygen concentration may be implicated (see sections Drug Interactions and Adverse Reactions).

Hepato-biliary disorders (see section Adverse Reactions):

Severe hepatocellular insufficiency may occur within the first 24 hours of IV amiodarone, and may sometimes be fatal. Close monitoring of transaminases is therefore recommended as soon as amiodarone is started.

Transplantation:

In retrospective studies, amiodarone use in the transplant recipient prior to heart transplant has been associated with an increased risk of primary graft dysfunction (PGD).

PGD is a life-threatening complication of heart transplantation that presents as left, right or biventricular dysfunction occurring within the first 24 hours of transplant surgery for which there is no identifiable secondary cause (see section Adverse Reactions). Severe PGD may be irreversible. For patients who are on the heart transplant waiting list, consideration should be given to use an alternative antiarrhythmic drug as early as possible before transplant.

Pregnancy and Lactation:

Pregnancy

There are insufficient data on the use of amiodarone during pregnancy in humans to judge any possible toxicity. However, in view of its effect on the foetal thyroid gland, amiodarone is contraindicated during pregnancy, except in exceptional circumstances.

Lactation

Amiodarone is excreted into the breast milk in significant quantities and breast-feeding is contra-indicated.

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Drug Interactions:
Some of the more important drugs that interact with amiodarone include warfarin, digoxin, phenytoin and any drug which prolongs the QT interval. Amiodarone raises the plasma concentrations of oral anticoagulants (warfarin) and phenytoin by inhibition of CYP 2C9. The dose of warfarin should be reduced accordingly. More frequent monitoring of prothrombin time both during and after amiodarone treatment is recommended. Phenytoin dosage should be reduced if signs of overdosage appear, and plasma levels may be measured.

Administration of Amidarone Injection to a patient already receiving digoxin will bring about an increase in the plasma digoxin concentration and thus precipitate symptoms and signs associated with high digoxin levels. Clinical, ECG and biological monitoring is recommended and digoxin dosage should be halved. A synergistic effect on heart rate and atrioventricular conduction is also possible.

Combined therapy with the following drugs which prolong the QT interval is contra-indicated (see section *Contraindications*) due to the increased risk of torsades de pointes; for example:

- Class Ia anti-arrhythmic drugs e.g. quinidine, procainamide, disopyramide.
- Class III anti-arrhythmic drugs e.g. sotalol, bretylium.
- Intravenous erythromycin, co-trimoxazole or pentamidine injection.
- Some anti-psychotics e.g. chlorpromazine, thioridazine, fluphenazine, pimozide, haloperidol, amisulpiride and sertindole.
- Lithium and tricyclic anti-depressants e.g. doxepin, maprotiline, amitriptyline.
- Certain antihistamines e.g. terfenadine, astemizole, mizolastine.
- Anti-malarials e.g. quinine, mefloquine, chloroquine, halofantrine.
- Moxifloxacin.

Fluoroquinolones
There have been rare reports of QTc interval prolongation, with or without torsades de pointes, in patients taking amiodarone with fluoroquinolones. Concomitant use of amiodarone with fluoroquinolones should be avoided (concomitant use with moxifloxacin is contra-indicated, see above).

Combined therapy with the following drugs is not recommended:

- Beta blockers and certain calcium channel inhibitors (diltiazem, verapamil); potentiation of negative chronotropic properties and conduction slowing effects may occur.
- Stimulant laxatives, which may cause hypokalaemia thus increasing the risk of torsades de pointes; other types of laxatives should be used.

Caution should be exercised over combined therapy with the following drugs which may also cause hypokalaemia and/or hypomagnesaemia, e.g. diuretics, systemic corticosteroids, tetracosactide, intravenous amphotericin.

In cases of hypokalaemia, corrective action should be taken and QT interval monitored. In case of torsades de pointes antiarrhythmic agents should not be given; pacing may be instituted and IV magnesium may be used.

Caution is advised in patients undergoing general anaesthesia, or receiving high dose oxygen therapy. Potentially severe complications have been reported in patients taking amiodarone undergoing general anaesthesia: bradycardia unresponsive to atropine, hypotension, disturbances of conduction, decreased cardiac output.

A few cases of adult respiratory distress syndrome, most often in the period immediately after surgery, have been observed. A possible interaction with a high oxygen concentration may be implicated.

Grapefruit juice inhibits cytochrome P450 3A4 and may increase the plasma concentration of amiodarone. Grapefruit juice should be avoided during treatment with amiodarone.

Drugs metabolised by cytochrome P450 3A4
When drugs are co-administered with amiodarone, an inhibitor of CYP 3A4, this may result in a higher level of their plasma concentrations, which may lead to a possible increase in their toxicity:

- Ciclosporin: plasma levels of ciclosporin may increase as much as 2-fold when used in combination. A reduction in the dose of ciclosporin may be necessary to maintain the plasma concentration within the therapeutic range.
- Statins: the risk of muscular toxicity is increased by concomitant administration of amiodarone with statins metabolised by CYP 3A4 such as simvastatin, atorvastatin and lovastatin. It is recommended to use a statin not metabolised by CYP 3A4 when given with amiodarone.
- Other drugs metabolised by cytochrome P450 3A4: examples of such drugs are lidocaine, tacrolimus, sildenafil, fentanyl, midazolam, triazolam, dihydroergotamine and ergotamine.

Flecainide
Given that flecainide is mainly metabolised by CYP 2D6, by inhibiting this isoenzyme, amiodarone may increase flecainide plasma levels; it is advised to reduce the flecainide dose by 50% and to monitor the patient closely for adverse effects. Monitoring of flecainide plasma levels is strongly recommended in such circumstances.

Interaction with substrates of other CYP 450 isoenzymes
In vitro studies show that amiodarone also has the potential to inhibit CYP 1A2, CYP 2C19 and CYP 2D6 through its main metabolite. When co-administered, amiodarone would be expected to increase the plasma concentration of drugs whose metabolism is dependent upon CYP 1A2, CYP 2C19 and CYP 2D6.

Incompatibilities:
Amiodarone Injection is incompatible with saline and should be administered solely in 5% dextrose solution. Amiodarone Injection, diluted with 5% dextrose solution to a concentration of less than 0.6mg/ml, is unstable. Solutions containing less than 2 ampoules Amidarone Injection in 500ml dextrose 5% are unstable and should not be used.

The use of administration equipment or devices containing plasticizers such as DEHP (di-2-ethylhexylphthalate) in the presence of amiodarone may result in leaching out of DEHP. In order to minimise patient exposure to DEHP, the final amiodarone dilution for infusion should preferably be administered through non DEHP-containing sets.

Adverse Reactions:
The following adverse reactions are classified by system organ class and ranked under heading of frequency using the following convention: very common (≥ 10%), common (≥ 1% and < 10%); uncommon (≥ 0.1% and < 1%); rare (≥ 0.01% and < 0.1%), very rare (< 0.01%), not known (cannot be estimated from the available data).

Blood and lymphatic system disorders:

- In patients taking amiodarone there have been incidental findings of bone marrow granulomas. The clinical significance of this is unknown.

Cardiac disorders:

- Common: bradycardia, generally moderate.
- Very rare:
 - marked bradycardia, sinus arrest requiring discontinuation of amiodarone, especially in patients with sinus node dysfunction and/or in elderly patients;
 - onset of worsening of arrhythmia, sometimes followed by cardiac arrest (see sections *Warnings & Precautions and Drug Interactions*).

Gastrointestinal disorders:

- Very rare: nausea.

General disorders and administration site conditions:

- Common: injection site reactions such as pain, erythema, oedema, necrosis, extravasation, infiltration, inflammation, induration, thrombophlebitis, phlebitis, cellulitis, infection, pigmentation changes.

Hepato-biliary disorders:

- Very rare:
 - isolated increase in serum transaminases, which is usually moderate (1.5 to 3 times normal range) at the beginning of therapy. They may return to normal with dose reduction or even spontaneously;
 - acute liver disorders with high serum transaminases and/or jaundice, including hepatic failure, sometimes fatal (see section *Warnings & Precautions*).

Immune system disorders:

- Very rare: anaphylactic shock.

Angioedema (there have been some reports of angioedema, although exact frequencies are not known).

Nervous system disorders:

- Very rare: benign intra-cranial hypertension (pseudo tumor cerebri), headache.

Respiratory, thoracic and mediastinal disorders:

- Very rare:
 - interstitial pneumonitis (see section *Warnings & Precautions*)
 - severe respiratory complications (adult acute respiratory distress syndrome), sometimes fatal (see sections *Warnings & Precautions and Drug Interactions*).
 - bronchospasm and/or apnoea in case of severe respiratory failure, and especially in asthmatic patients.

Skin and subcutaneous tissue disorders:

- Very rare: sweating.
- Not known: urticaria.

Vascular disorders:

- Common: decrease in blood pressure, usually moderate and transient. Cases of hypotension or collapse have been reported following overdosage or a too rapid injection.
- Very rare: hot flushes.

Injury, poisoning and procedural complications:

- Frequency not known: potentially fatal primary graft dysfunction post cardiac transplant (see section *Warnings and Precautions*).

Overdosage:
There is no information regarding overdosage with intravenous amiodarone. Little information is available regarding acute overdosage with oral amiodarone. Few cases of sinus bradycardia, heart block, attacks of ventricular tachycardia, torsades de pointes, circulatory failure and hepatic injury have been reported.

In the event of overdose, treatment should be symptomatic, in addition to general supportive measures. The patient should be monitored and if bradycardia occurs beta-adrenostimulants or glucagon may be given.

Spontaneously resolving attacks of ventricular tachycardia may also occur. Due to the pharmacokinetics of amiodarone, adequate and prolonged surveillance of the patient, particularly cardiac status, is recommended.

Neither amiodarone nor its metabolites are dialysable.

Dosage Form and Packaging Available:
Amiodarone HCl 150mg/3 mL Injection Bioindustria L.I.M. is available in 3mL glass ampoule packed into a carton box. Each carton box consists of 5 ampoules and one product insert.

Cautions for Usage:
Because of the pharmaceutical characteristics, concentration less than 600mg/L should not be used (see also section *Incompatibilities*). Only 5% dextrose should be used. When the product is mixed with diluent solution (1 ampoule in 250ml of 5% dextrose), the mixture is clear and free of particles. Do not mix with other preparations in infusion solution.

Storage Conditions:
Store at room temperature below 30°C, protect from light.
Discard any unused portion after dilution.
Do not refrigerate or freeze.
Storage at low temperatures (below 15°C) could cause a formation of crystalline precipitate.
The ampoule in which is present a precipitate should not be used.

Manufacturer: Bioindustria Laboratorio Italiano Medicinali S.p.A., Via De Ambrosiis 2, 15067 Novi Ligure (AL), Italy.

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